

Sound2Haptic: A Toolkit for Portable Multi-Channel Haptic Integration Across Multiple Form Factors and Devices

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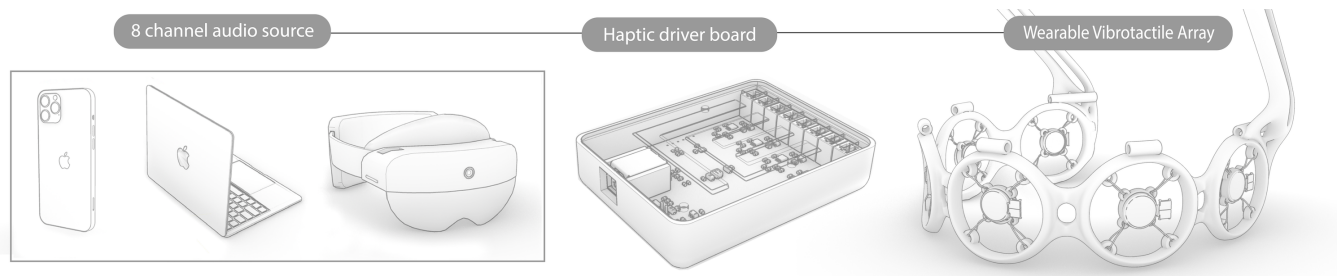


Figure 1: An overview of the Sound2Haptic system.

Abstract

Existing multi-actuator vibrotactile systems often require external hardware such as sound cards and haptic amplifiers, which limits portability and creates complexity for non-technical users. This presents a significant barrier for researchers and designers in fields like human factors and healthcare. We present Sound2Haptic, an vibrotactile toolkit that integrates a sound card and haptic amplifiers into a single device. The toolkit connects to laptops, phones, and XR headsets, enabling portable eight-channel multi-actuator interaction accessible to non-technical users. The toolkit features a novel mechanical design that reduces cross-actuator interference and enables form factor customization. We demonstrate the toolkit's functional efficacy through psychophysical evaluation across three form factors, and its ease of use through three case studies: (1) a clinical application for tinnitus research (2) a human factors study

on speech prosody conducted with human factors researcher, and (3) an exploration of spatial neglect rehabilitation using XR and haptics.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Haptics, Vibrotactile, Accessibility, Toolkit, Actuators, Haptic interfaces, Vibrations

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1 Introduction

Haptics research often requires significant upfront investment in both hardware and software development, creating a substantial

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barrier to entry for researchers wanting to explore haptic applications. Existing multi-actuator vibrotactile systems require external hardware like sound cards and haptic amplifiers, limiting portability and creating complexity that makes them inaccessible to non-technical users. This demonstration introduces Sound2Haptic, an all-in-one vibrotactile toolkit that integrates sound cards and haptic amplifiers into a single device. Our system enables portable eight-channel multi-actuator interaction that connects to laptops, phones, and XR headsets, making research involving accessible to non-technical users. The toolkit features a novel mechanical design that reduces cross-actuator interference and includes customizable form factors, demonstrated through bracelet, necklace, and over-ear configurations.

We validated the toolkit through psychophysical evaluation, measuring vibrotactile perception capabilities across all form factors. Results show superior spatial localization for necklace and over-ear devices, while bracelet and necklace configurations demonstrated better vibration sensitivity. Three case studies demonstrate practical applications: tinnitus research, speech prosody studies, and spatial neglect rehabilitation, showcasing how Sound2Haptic enables haptic research without requiring extensive technical expertise.

2 Related Work: Vibrotactile Toolkits for Virtual Reality and Smartphones

Several vibrotactile toolkits have been proposed to work in virtual reality applications or with smartphones. An early pioneer, VITAKI, was an ERM-based toolkit targeting virtual reality and video game applications [4]. It used shift registers and PWM drivers to control up to sixteen ERMs. While it was shown effective in implementing an object grasping and falling interaction in VR, the custom software only supported simple wave design and could not reconstruct audio waveforms. Alternatively, the Stereohaptics toolkit used an editing workflow similar to audio pipelines, enabling users to create, record, and playback simple haptic waveforms, making haptic design accessible through familiar audio tools [10]. However, its audio processing kit only supported two actuators, limiting the design space of possible applications. Syntacts is a more sophisticated toolkit, where designers can compose complex waveforms in its GUI editor and send them to eight actuators for testing [5]. However, a bulky desktop audio amplifier is needed to support the vibration playback. This limitation in portability constrains its usage in wearable applications. In summary, no existing toolkit has integrated audio-based design workflows specifically for creating wearable vibrotactile feedback in XR and mobile contexts.

3 System Design

In this section, we provide a technical overview of the Sound2Haptic system, including the tactile hardware implementation, and the custom PCB driver. We demonstrate Sound2Haptic in three wearable form factors: bracelet (Figure 2A), necklace, and over-ear (Figure 2B and 2C). The bracelet and necklace devices wrap around the wrist and neck respectively, and are secured with adjustable clasps. The necklace and overear devices position the actuators on the back of the neck, as prior research has demonstrated the viability of the neck as a site for haptic feedback [8].

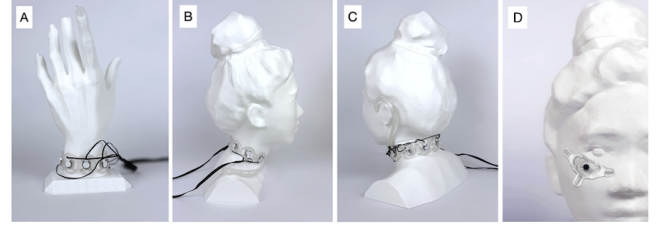


Figure 2: Wearable prototypes demonstrate the motor array’s customizability in shape and placement: (A) wristband, (B) necklace, (C) over-ear device, and (D) single-actuator device. Designs are 3D-printed and use suspended motors to reduce cross-actuator vibration.

3.1 Mechanical Design

The motor array frame is highly customizable, with flexibility in actuator number and device shape. The repeating module of an inner motor ring suspended within an outer motor ring can be arranged in any pattern. As an example, the number of motors in the necklace and bracelet devices can be configured in a parametric grasshopper model (Figure 4 shows a screen shot of the interface).

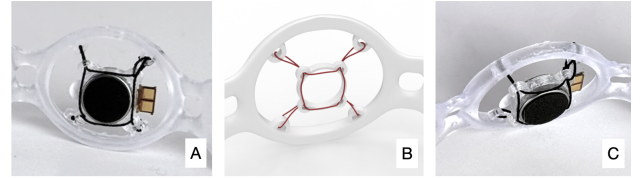


Figure 3: (A) LRA motor press-fit into inner motor ring. (B) Threading method used for suspension. (C) Side view showing inward motor protrusion for skin contact.

Each motor is secured by press fit (and optionally, glue) into an inner motor ring (Figure 3A), enabling efficient motor replacement, and device disassembly. Each inner motor ring is tethered to the main structure using a single piece of thread that is woven between the inner and outer rings at four suspension points. 3B) This suspension system has the advantage of allowing the motor to move freely in the horizontal direction with minimal attached material (Figure 3C), making it lightweight, quiet, and effective at preventing unwanted vibration spread.

The motor array frames are fabricated with accessible 3D-printing technology; The neck and wrist devices are SLA-printed with Flexible 80A Resin, and the over device is FDM printed with TPU.

3.2 Electrical Design

Our electrical (Figure 5) design integrates sound card and haptic driver functionality into a single custom PCB, eliminating the need for external sound cards required by prior systems like TAPS [7] and Syntacts [5]. We leverage existing audio libraries such as Python’s sounddevice[2], which wraps the commonly used PortAudio framework. PortAudio’s existing drivers recognize our device as

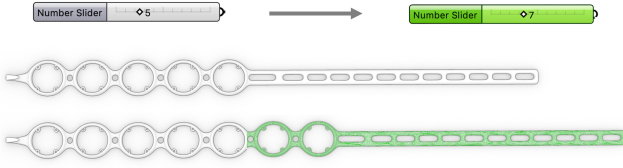


Figure 4: Parametric Grasshopper model enables customization of actuator count and layout. Adjusting a slider dynamically updates the 3D-printed motor array frame for custom necklace and bracelet designs.

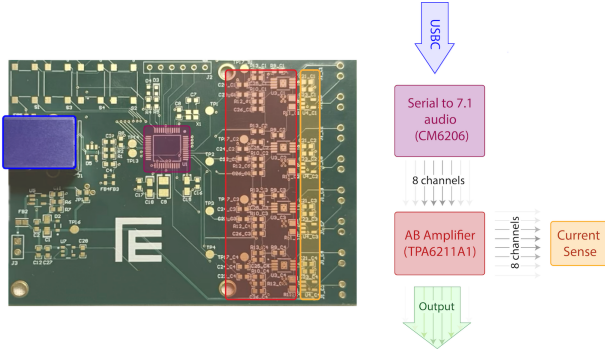


Figure 5: Custom PCB integrates USB sound card and 8-channel haptic amplifier system. Using the CM6206 audio chip and TPA6211A1 amplifiers, it supports standard audio libraries and adds real-time current sensing for device fitting, enabling seamless plug-and-play functionality.

an 8-channel sound card, enabling seamless compatibility with laptops, phones, and XR headsets like the HoloLens without requiring custom drivers or software installations.

The core of our design uses the CMedia CM6206 USB-to-7.1 audio chip, which converts serial audio output from standard PortAudio drivers into 8 channels of synchronous haptic signals. We pair this with TPA6211A1 amplifiers configured with standard headphone gain, inspired by Pezent et al.’s approach [5]. A key innovation is our simplified fit measurement system using current sensing rather than back-EMF techniques employed by prior work like VHP[1]. We implement this using the INA180 current sense amplifier with a test resistor circuit, allowing us to send test signals to assess device tightness. This approach is significantly simpler than back-EMF measurement because it doesn’t require switching motors to high-impedance mode, streamlining both the hardware design and measurement process.

4 Psychophysical Evaluation

To evaluate our design, we conducted a user study to verify that the actuators could be perceptually distinguished, and provide sufficient skin contact for low-amplitude perception. We measured amplitude thresholds and localization accuracy, comparing the wrist and neck-worn devices.

4.0.1 Methods. For the in-person psychophysical evaluation, 9 participants (mean age: 54, min age: 20, max age: 97) each completed a 75-minute session. Participants completed a pre-experiment questionnaire collecting demographic information including hearing loss history. Each participant tested all three form factors (necklace, bracelet, over-ear) in a random, counterbalanced order. To mask audio cues from mechanical vibration, participants wore in-ear headphones (Airpods Pro 2) playing a 170Hz tone during testing. Participants completed three discrimination tasks in random order: (1) Absolute threshold: minimum detectable amplitude, (2) Localization Accuracy: identifying which motor is active, and (3) Relative Localization: determining if sequential vibrations occur at same or adjacent locations. All tasks were conducted at 170 Hz, with conditions repeated across ten trials. Additional information about the test procedure can be found in supplemental information.

4.0.2 Results and Discussion. Performance metrics are summarized in Table 1. Wilcoxon tests showed bracelet and necklace were similar in Absolute Thresholds, while both differed significantly from overear. The neck’s superior performance aligns with bilateral symmetry enhancing spatial discrimination on midline regions [6]. Absolute Localization Accuracy varied significantly by device, with post-hoc Bonferroni tests showing bracelet performed worse than both necklace and overear. Relative Localization Accuracy showed similar patterns, with bracelet significantly worse than overear. Improvements in relative versus absolute localization suggest sequential stimuli reduce cognitive load [3], with motion illusions potentially further enhancing accuracy [9]. See supplemental information for more details.

Table 1: Performance metrics of different device form factors. Absolute Threshold is given in SoundDevice input level and peak acceleration in m/s^2 . Localization results are average percent accuracy.

Device	Threshold	Localization	Relative Loc.
Bracelet	0.096 (2.43 m/s^2)	0.528	0.753
Necklace	0.09 (2.38 m/s^2)	0.670	0.861
Overear	0.179 (3.79 m/s^2)	0.756	0.892

5 Case Studies and Proposed Demonstrations

To validate Sound2Haptic’s accessibility and utility for non-technical users, we conducted three case studies across diverse research domains, demonstrating how the toolkit enables rapid prototyping without requiring haptic engineering expertise. Some of these case studies will be featured as interactive demos during the conference.

5.1 Clinical Tinnitus Research

Our first case study involved a clinical tinnitus research application requiring synchronized visual, audio, and haptic signals. The researchers had initially attempted building their own prototype using a Class D amplifier, but encountered issues with excessive noise and heat generation. They needed a single-actuator configuration. By integrating Sound2Haptic into their existing experimental setup, they successfully achieved the precise multi-modal synchronization

required for their clinical studies. Here, Sound2Haptic allowed them to focus on their research rather than technical implementation challenges.

5.2 Speech Prosody Human Factors

The second case study featured a human factors researcher with a design background investigating haptics for speech prosody perception. She successfully integrated our device into a web-based experimental platform and conducted user studies examining how haptic feedback augments speech understanding. This implementation required minimal technical setup, as Sound2Haptic is compatible with standard web technologies, and is plug-and-play in nature.

5.3 Spatial Neglect Workshop Exploration

Our third case study explored spatial neglect rehabilitation through XR and haptics in a workshop setting. Spatial neglect, where patients lose awareness of parts of their sensory field, has shown improvement with haptic intervention. We developed a demonstration based on existing audio-visual rehabilitation approaches for stroke patients to illustrate how Sound2Haptic facilitates rapid prototyping of applications combining XR and haptic feedback.

6 Conclusion

Sound2Haptic addresses critical barriers in vibrotactile research by integrating sound card and haptic amplifier functionality into a single, portable device. Our toolkit enables non-technical users to rapidly prototype multi-channel haptic applications across diverse domains, from clinical tinnitus research to spatial neglect rehabilitation. The modular mechanical design with suspension-based vibration isolation, combined with plug-and-play compatibility across laptops, phones, and XR headsets, democratizes access to sophisticated haptic research tools. Through psychophysical validation and three case studies, we demonstrate that Sound2Haptic successfully bridges the gap between complex haptic engineering and accessible research prototyping, enabling broader adoption of haptic technologies in human factors and healthcare applications.

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